$\underline{\text{Extra}}$: $my''+by'+ky=f(t)$ is underdamped $b^2-4mk<0$

suppose $y(0)=0, y'(0)=0$. $|f(t)|\le A$

so solution $y = h(t) * f(t)$

$$h(t) = \mathcal{L}^{-1}\{H(s)\} = \mathcal{L}^{-1}\left\{\frac{1}{ms^2+bs+k}\right\}$$

$$= \mathcal{L}^{-1}\left\{\frac{1}{m(s^2+\frac{b}{m}s+\frac{k}{m})}\right\} \quad \pm\left(\frac{b}{2m}\right)^2$$

$$= \mathcal{L}^{-1}\left\{\frac{1}{m\left((s+\frac{b}{2m})^2+\frac{k}{m}-\frac{b^2}{4m^2}\right)}\right\}$$

$$= \mathcal{L}^{-1}\left\{\frac{1}{m}\cdot\frac{1}{(s+\frac{b}{2m})^2+\frac{4mk-b^2}{4m^2}}\right\} \quad 4mk-b^2>0$$

$$\sqrt{\frac{4mk-b^2}{4m^2}} = \frac{\sqrt{4mk-b^2}}{2m}$$

$$= \mathcal{L}^{-1}\left\{\frac{1}{m}\cdot\frac{2m}{\sqrt{4mk-b^2}}\cdot\frac{\frac{\sqrt{4mk-b^2}}{2m}}{(s+\frac{b}{2m})^2+\frac{4mk-b^2}{4m^2}}\right\}$$

$$= \frac{2}{\sqrt{4mk-b^2}}\cdot e^{-\frac{b}{2m}t}\sin\left(\underbrace{\frac{\sqrt{4mk-b^2}}{2m}}_{\beta}t\right) = \frac{1}{m\beta}e^{-\frac{b}{2m}t}\sin(\beta t)$$

so $$y = h(t)*f(t) = \overset{f(t)*h(t)}{} \int_0^t \underset{\le A}{f(t-v)}\frac{1}{m\beta}e^{-\frac{b}{2m}v}\sin(\beta v)\,dv$$

as $|f(t)|\le A$ $\Rightarrow |y(t)| \le \int_0^t A\frac{1}{m\beta}e^{-\frac{b}{2m}v}|\sin(\beta v)|\,dv$

$\underline{\text{Note}}$: $|\sin(\beta v)|\le 1$

$$\Rightarrow |y(t)| \leq \int_0^t A \frac{1}{m\beta} e^{-\frac{b}{2m}v} \cdot 1 \, dv$$

$$= A \frac{1}{m\beta} \cdot \frac{1}{-\frac{b}{2m}} e^{-\frac{b}{2m}v} \Big|_0^t$$

$$\Rightarrow |y(t)| \leq \frac{A}{m\beta}\left(-\frac{2m}{b} e^{-\frac{b}{2m}t} - \left(-\frac{2m}{b} e^{0}\right)\right)$$

$$= \frac{A}{m\beta}\Big(\underbrace{-\frac{2m}{b} e^{-\frac{b}{2m}t}}_{\text{negative}} + \frac{2m}{b} \Big)$$

$$\leq \frac{A}{m\beta}\Big(\overset{\rightarrow 0}{0} + \frac{2m}{b} \Big) = \frac{2A}{b\beta}$$

If $|f(t)| \leq A \Rightarrow |y(t)| \leq \frac{2A}{b\beta}$

Ex: Suppose $m = 5\,kg$, $k = 3000\ N/m$, $A = 10\,N$

$$\beta = \frac{\sqrt{4mk - b^2}}{2m} = \frac{\sqrt{4(5)(3000) - b^2}}{10} = \frac{\sqrt{60000 - b^2}}{10}$$

If we want $|y(t)| \leq 0.01 \Rightarrow \underset{|y(t)| \leq}{\frac{2A}{b\beta}} \leq 0.01$

$$\frac{2 \cdot 10}{b\frac{\sqrt{60000 - b^2}}{10}} \leq 0.01 \Rightarrow \frac{200}{b\sqrt{60000 - b^2}} \leq 0.01$$

$$\Rightarrow b\sqrt{60000 - b^2} \geq 20000$$

$$\Rightarrow b^2(60000 - b^2) \geq (20000)^2$$

$$= \frac{A}{\beta m}\left[\frac{1}{-\frac{b}{2m}} e^{-\frac{b}{2m}v}\Big|_{v=0}^{v=t}\right]$$

$$= \frac{A}{\beta m}\Big[\underbrace{-\frac{2m}{b} e^{-\frac{bt}{2m}}}_{<0 \text{ (negative)}} + \frac{2m}{b}\cdot 1\Big] \le \frac{A}{\beta m}\left[0 + \frac{2m}{b}\right]$$

$$= \frac{2A}{\beta b}$$

Therefore $|y(t)| \le \frac{At}{\beta m}$, $|y(t)| \le \frac{2A}{\beta b}$

<u>Ex</u>: Suppose $m = 5$ kg, $k = 3000$ N/m, $A = 10$ N

then $\beta = \sqrt{\frac{4mk - b^2}{4m^2}} = \sqrt{\frac{60000 - b^2}{100}} = \frac{\sqrt{60{,}000 - b^2}}{10}$

If we want $|y(t)| \le 0.01$ m $\Rightarrow \frac{2A}{\beta b} \le 0.01$

$$\Rightarrow \frac{20}{\frac{\sqrt{60000 - b^2}}{10}\, b} \le 0.01 \Rightarrow \frac{200}{b\sqrt{60000 - b^2}} \le 0.01$$

$$\Rightarrow 20000 \le b\sqrt{60000 - b^2}$$

$$\Rightarrow 400000000 \le b^2(60000 - b^2)$$

$$\Rightarrow b^4 - 60000 b^2 + (20000)^2 \le 0$$

$$\Rightarrow b^2 = \frac{60000 \pm \sqrt{(60000)^2 - 4(20000)^2}}{2} \Rightarrow b = \sqrt{\frac{60000 \pm 20000\sqrt{5}}{2}}$$

$$\Rightarrow b = 100\sqrt{3 \pm \sqrt{5}}$$

So for $100\sqrt{3 - \sqrt{5}} \le b \le 100\sqrt{3 + \sqrt{5}}$

$$\Rightarrow b^4 - 60000b^2 + (20000)^2 \leq 0$$

$$\Rightarrow b^2 = \frac{60000 \pm \sqrt{(60000)^2 - 4(20000)^2}}{2}$$

$\vdots$

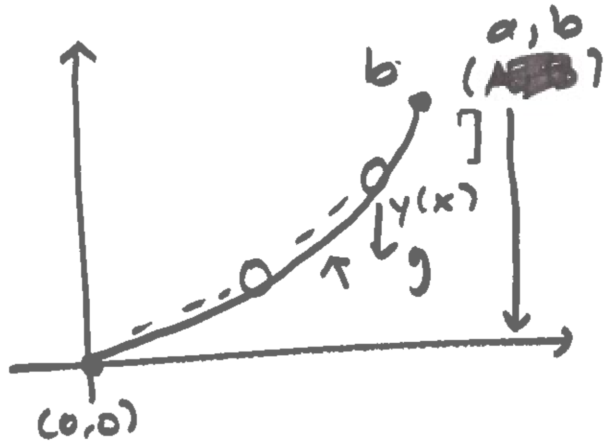
$$\Rightarrow b = 100\sqrt{3 \pm \sqrt{5}} \quad \begin{array}{l} \nearrow \approx 87.4032 \\ \rightarrow \approx 228.82456 \end{array}$$

we want $87.4032 \leq b \leq 228.82456$

guarantees $|y(t)| \leq 0.01$

Ex: Suppose you want to construct a pendulum

same amount of time.



$T(y)$: time to "fall" to $(0,0)$.

$T(b) = T_0$

arc length

Call $f(y) = \dfrac{ds}{dy} = \sqrt{1+\left(\dfrac{dx}{dy}\right)^2}$

arclength $= \displaystyle\int_a^b \sqrt{1+(y')^2}\,dx$

y variable $s = \displaystyle\int_c^d \sqrt{1+\left(\dfrac{dx}{dy}\right)^2}\,dy$

Conservation of Energy

kinetic energy : $\frac{1}{2}mv^2$

$= \frac{1}{2}m\left(\frac{ds}{dt}\right)^2$

change in potential energy :

$$mgb - mgy = mg(b-y)$$

$$\Rightarrow \frac{1}{2}m\left(\frac{ds}{dt}\right)^2 = mg(b-y)$$

$$\Rightarrow \frac{ds}{dt} = \pm\sqrt{2g(b-y)} = -\sqrt{2g(b-y)}$$

$$\Rightarrow \int_b^0 dt = -\frac{1}{\sqrt{2g(b-y)}}\,ds = \int_b^0 -\frac{1}{\sqrt{2g(b-y)}}\frac{ds}{dy}\,dy$$

$$T(b) = \int_b^{a\ b} dt = \int_b^0 -\frac{1}{\sqrt{2g(b-y)}}\frac{ds}{dy}\,dy$$

$$\Rightarrow \underset{T_0}{\underset{\parallel}{T(b)}} = \int_0^b \frac{1}{\sqrt{2g(b-y)}} f(y)\,dy \leftarrow \text{convolution}$$

$$\longrightarrow \frac{1}{\sqrt{b-y}} f(y)$$

$$\Rightarrow T_0 = \frac{1}{\sqrt{2g}} \left(y^{-1/2} * f(y) \right)$$

$$\Rightarrow \mathcal{L}\left\{ T_0 = \frac{1}{\sqrt{2g}} \left(y^{-1/2} * f(y) \right) \right\}$$

$$\Rightarrow \frac{T_0}{s} = \frac{1}{\sqrt{2g}} \mathcal{L}\{y^{-1/2}\} \cdot F(s)$$

$$= \frac{1}{\sqrt{2g}} \cdot \frac{\Gamma(1/2)}{s^{1/2}} F(s)$$

$$\Rightarrow \frac{T_0}{s} = \frac{1}{\sqrt{2g}} \frac{\sqrt{\pi}}{\sqrt{s}} F(s)$$

$$\Rightarrow F(s) = \frac{T_0\sqrt{2g}}{\sqrt{\pi}\sqrt{s}} = \frac{T_0\sqrt{2g}}{\sqrt{\pi}} \frac{1}{\sqrt{s}}$$

$\sqrt{\pi}$

$\sqrt{\pi}$

$$\Rightarrow F(s) = \frac{T_0\sqrt{2g}}{\pi} \frac{\sqrt{\pi}}{\sqrt{s}} \Rightarrow f(y) = \frac{B\sqrt{2g}}{\pi} y^{-1/2}$$

$$\Rightarrow \frac{ds}{dy} = \frac{T_0\sqrt{2g}}{\pi} y^{-1/2}$$

$$\Rightarrow \sqrt{1 + \left(\frac{dx}{dy}\right)^2} = \frac{ds}{dy} = \frac{T_0\sqrt{2g}}{\pi} y^{-1/2}$$

$$\Rightarrow 1 + \left(\frac{dx}{dy}\right)^2 = \underbrace{\boxed{\frac{2g\,T_0^2}{\pi^2}}}_{r} y^{-1} \cdots \rightarrow \boxed{\frac{dx}{dy} = \sqrt{\frac{2r-y}{y}} \text{ (separable)}}$$

sub: $y = 2r\sin^2\left(\frac{\theta}{2}\right)$

$$\mathcal{L}\{t^n\} = \frac{n!}{s^{n+1}}$$

$$\mathcal{L}\{t^r\} = \int_0^\infty e^{-st} t^r\,dt$$

$$= \frac{1}{s} \underbrace{\int_0^\infty e^{-st} t^{r-1}\,dt}$$

$$= \frac{\Gamma(r+1)}{s^{r+1}}$$

$\Gamma(n) = (n+1)!$ n pos integer

$$\Gamma(1/2) = \sqrt{\pi}$$

$$\mathcal{L}\{t^n\} = \frac{\Gamma(n+1)}{s^{n+1}}$$

$$\mathcal{L}\{t^{-1/2}\} = \frac{\Gamma(1/2)}{s^{1/2}} = \sqrt{\frac{\pi}{s}}$$

$$\Gamma(1/2) = \sqrt{\pi}$$

So $T_0 = \frac{1}{\sqrt{2g}}\left(y^{-1/2} * f(y) \right)$

$$\Rightarrow \frac{T_0}{s} = \frac{1}{\sqrt{2g}}\left(\mathcal{L}\{y^{-1/2}\} F(s) \right)$$

$$\Rightarrow \frac{T_0}{s} = \frac{1}{\sqrt{2g}}\left(\sqrt{\frac{\pi}{s}}\, F(s) \right)$$

$$\Rightarrow F(s) = \sqrt{\frac{2g}{\pi}} \cdot \frac{T_0}{\sqrt{s}} = \frac{\sqrt{2g}}{\pi} T_0 \frac{\sqrt{\pi}}{\sqrt{s}}$$

$$\Rightarrow f(y) = \frac{\sqrt{2g}}{\pi} T_0\, y^{-1/2} \Rightarrow \frac{ds}{dy} = \frac{\sqrt{2g}}{\pi} T_0\, y^{-1/2}$$

$$\left(\frac{dx}{dy}\right)^2 + 1 = \left(\frac{ds}{dy}\right)^2 = \frac{2g}{\pi^2} T_0^2\, y^{-1} = 2r\,\frac{1}{y} \qquad r = \frac{g T_0^2}{\pi^2}$$

$$\Rightarrow \left(\frac{dx}{dy}\right)^2 + 1 = \frac{2r}{y}$$

$$\Rightarrow \left(\frac{dx}{dy}\right)^2 = \frac{2r}{y} - 1 = \frac{2r - y}{y}$$

$$\Rightarrow \frac{dx}{dy} = \sqrt{\frac{2r-y}{y}} \Rightarrow dx = \sqrt{\frac{2r-y}{y}}\, dy \quad \text{(separable)}$$

using $y = 2r\sin^2\left(\frac{\theta}{2}\right) \Rightarrow dx = r(\cos\theta + 1)\, d\theta$

$$\Rightarrow x = r(\theta + \sin\theta)$$

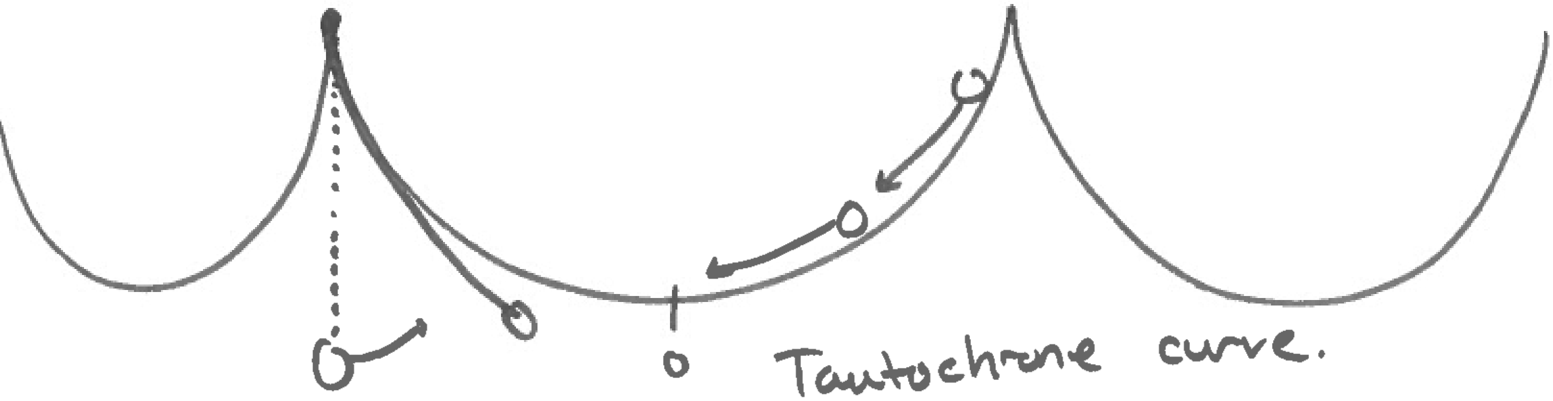
since $y = 2r\sin^2\left(\frac{\theta}{2}\right) = r(1 - \cos\theta)$

we get
$$\begin{cases} x = r(\theta + \sin\theta) \\ y = r(1 - \cos\theta) \end{cases}$$

$$\cdots \to \quad dx = r(\cos\theta + 1)\, d\theta$$

$$\Rightarrow \begin{cases} x = r(\sin\theta + \theta) \\ y = r(1 - \cos\theta) \end{cases}$$

cycloid



Tautochrone curve.

cycloid

brachistochrone curve